Groups, rings and modules

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Contents

Lecture 1 – 22/01/26

§1 Groups – a reminder

Definition 1.1. A *group* is a triple $(G, \cdot, e)$ where G is a set,

$$\cdot : G \times G \to G$$

is a function, and $e \in G$ such that

- *Associativity.* $(a \cdot b) \cdot c = a \cdot (b \cdot c)$;
- *Identity.* For all $a \in G$, $a \cdot e = e \cdot a = a$;
- *Inverses.* For all $a \in G$, there exists $a^{-1} \in G$ such that $a \cdot a^{-1} = a^{-1} \cdot a = e$.

The size of G, $|G|$, is called the *order* of the group.

Proposition 1.2 (Uniqueness of inverses)
Inverses are unique.

Proof. Left as an exercise (see 1A). ■

Definition 1.3 (Subgroups)**.** If G is a group and $H \subseteq G$, then we say H is a *subgroup* if

- $e \in H$;
- for all a, $b \in H$, $a \cdot b \in H$;
- $(H, \cdot, e)$ is a group.

In order to say item 3, we must first establish 1 and 2.

Proposition 1.4

A non-empty subset $H \subseteq G$ is a subgroup if and only if, for all h_1, $h_2 \in H$, $h_1 h_2^{-1} \in H$.

Proof. Left as an exercise (see 1A). ■

Definition 1.5. A group G is called *abelian* if, for all g_1, $g_2 \in G$, $g_1 g_2 = g_2 g_1$.

Example 1.6

- We have $(\mathbb{Z}, +)$, $(\mathbb{Q}, +)$, $\mathbb{R}$, $\mathbb{C}$, $\mathbb{Z}/n\mathbb{Z}$.
- The symmetric groups S_n: permutations of $\{1, \ldots, n\}$ under composition. More generally for any set X, we have $\mathrm{Sym}(X)$.
- The dihedral groups D_{2n}: the group of symmetries of a regular n-gon.
- The general linear group over $\mathbb{R}$, $\mathrm{GL}(n, \mathbb{R})$ or $\mathrm{GL}_n(\mathbb{R})$, is the set of $\{f : \mathbb{R}^n \to \mathbb{R}^n \mid f \text{ is an } \mathbb{R}\text{-linear bijection}\}$. Since $\mathrm{GL}(n, \mathbb{R})$ is made of functions, we can think of it a subset of $\mathrm{Sym}(\mathbb{R}^n)$. (Equivalently, it's the set of n by n invertible matrices with elements in $\mathbb{R}$).

Some examples of subgroups:

- The even permutations $A_n \leq S_n$ (the alternating group);
- Rotational symmetries of an n-goin, $C_n \leq D_{2n}$;
- Determinant one matrices $\mathrm{SL}(n, \mathbb{R}) \leq \mathrm{GL}(n, \mathbb{R})$.

Other examples include $C_2 \times C_2$, $Q_8 = \{\pm 1, \pm i, \pm j, \pm k\}$.

Definition 1.7. Let G be a group, $g \in G$ and $H \leq G$ a subgroup. Then the *left coset*

$$gH = \{gh : h \in H\}$$

Basic properties include that $g \in gH$, so cosets are non-empty. Also, cosets partition G – they have no non-trivial overlaps and every g lies in some coset. Furthermore, the map

$$\begin{aligned} H &\to gH \\ h &\mapsto gh \end{aligned}$$

defines a bijection.

Theorem 1.8 (Lagrange)

If G is a finite group and $H \leq G$ then

$$|G| = |H| \cdot |G : H|$$

where $|G : H|$, the index of H in G, is the number of cosets of H in $|G|$.

Definition 1.9. Let G be a group and let $g \in G$. The *order* of g, $\mathrm{ord}(g)$, is the smallest positive integer n such that $g^n = e$, or infinity if no such n exists.

Proposition 1.10

Let $g \in G$. Then, $\mathrm{ord}(g) \mid |G|$.

Proof. Let $g \in G$ and consider $H = \{e, g, g^2, \dots, g^{n-1}\}$ where $n = \mathrm{ord}(g)$. This is a subgroup: it is non-empty, and $g^r \cdot g^{-s} = g^{r-s} \in H$ (after potentially multiplying by $g^n = e$). Also, the elements of H are distinct – if $g^i = g^j$ for $0 \leq i < j \leq n$, then $g^{j-i} = e$, contradicting the minimality of $\mathrm{ord}(g)$. Hence $|H| = ord(g)$ – now apply Lagrange. ∎

§2 Normal subgroups, quotients and homomorphisms

Let $H \leq G$. When are two cosets of H in G equal? If $gH = g'H$, then $g \in g'H$, so $g = g'h$ for some $h \in H$. In other words, $gH = g'H \iff (g')^{-1}g \in H$.

We denote the set of left cosets of $H \leq G$ by $G/H = \{gH : g \in G\}$. Does G/H carry a natural group structure?

Lecture 2 – 24/01/26

We propose the 'natural' group structure:

$$g_1H \cdot g_2H = (g_1g_2)H$$

But this rule may depend on the coset representatives we pick. We must check that it doesn't. Suppose $g_2H = g_2'H$, so $g_2' = g_2h$ for some $h \in H$. Then

$$g_1H \cdot g_2'H = (g_1g_2')H = g_1g_2hH = g_1g_2H$$

On the other hand, if $g_1H = g_1'H$, so $g_1' = g_1h$ for $h \in H$. Then

$$g_1'H \cdot g_2H = g_1hg_2H$$

and that is not automatically equal to g_1g_2H – we need $g_2^{-1}hg_2 \in H$ to hold for all $g_2 \in G$, $h \in H$.

Definition 2.1 (Normal subgroups)**.** A subgroup $H \leq G$ is *normal* if, for all $g \in G$, $h \in H$, $ghg^{-1} \in H$. We write this as $H \triangleleft G$.

> **Remark 2.2.** When $H \triangleleft G$, then G/H inherits a group structure by the rule above. We will refer to this as the *quotient group of* G *by* H.

Definition 2.3 (Homomorphisms)**.** Let G and H be groups. A homomorphism from G to H is a function $f : G \to H$ such that, for all $g_1, g_2 \in G$, $f(g_1g_2) = f(g_1)f(g_2)$.

Results such as $f(e_G) = e_H$ follow immediately.

Lemma 2.4

If $f : G \to H$ is a homomorphism, then $f(g^{-1}) = f(g)^{-1}$.

Proof. Calculate $f(g \cdot g^{-1})$ in two ways.

① $f(gg^{-1}) = f(e_G) = e_H$;

② $f(gg^{-1}) = f(g)f(g^{-1})$

so $f(g)f(g^{-1}) = e_H$ and the result follows. ∎

Definition 2.5. Let $f : G \to H$ be a homomorphism. The *kernel* of f is

$$\ker f = \{g \in G : f(g) = e_H\}$$

and the *image* of f is

$$\operatorname{Im} f = \{h \in H : h = f(g) \text{ for some } g\}$$

Proposition 2.6

Let $f : G \to H$ be a homomorphism. Then $\ker f \triangleleft G$ and $\operatorname{Im} f \leq H$.

Proof. Observe $e \in \ker f$ so $\ker f$ non-empty. Now, if $x, y \in \ker f$, then

$$f(xy^{-1}) = f(x)f(y)^{-1} = e$$

so $xy^{-1} \in \ker f$. For normality, let $x \in G$ and $g \in \ker f$. Now

$$f(xgx^{-1}) = f(x)ef(x)^{-1} = e$$

so $xgx^{-1} \in \ker f$ and hence $\ker f \triangleleft G$.

For the image, observe that, if $a, b \in \operatorname{Im} f$ with $a = f(x)$, $b = f(y)$, then $ab^{-1} = f(xy^{-1}) \in \operatorname{Im} f$. ∎

> **Remark.** Observe $f : G \to G/H$ taking $g \mapsto gH$ is a homomorphism. This is a nice way of expressing its 'naturalness'.

Definition 2.7. A homomorphism $f : G \to H$ is an *isomorphism* if it is a bijection. Two groups G and H are said to be *isomorphic*, $G \cong H$, if there exists an isomorphism between them.

Theorem 2.8 (First isomorphism theorem)

Let $f : G \to H$ be a homomorphism, then $\ker f$ is normal in G and the homomorphism

$$\begin{aligned} G/\ker f &\to \operatorname{Im} f \\ g \ker f &\mapsto f(g) \end{aligned}$$

is an isomorphism. In particular, $G/\ker f \cong \operatorname{Im} f$.

Proof. We have already shown the normality result. Consider the map

$$\begin{aligned}\phi : G/\ker f &\to \operatorname{Im} f \\ g\ker f &\mapsto f(g)\end{aligned}$$

- *ϕ is well-defined.* If $g\ker f = g'\ker f$, then $g'g^{-1} \in \ker f$, so $f(g'g^{-1}) = e$ which gives $f(g) = f(g')$ as required.
- *ϕ is surjective.* Let $h \in \operatorname{Im} f$. Then $h = f(g)$ for some $g \in G$, and so $h = \phi(g\ker f)$.
- *ϕ is injective.* Suppose $\phi(g\ker f) = \phi(g'\ker f)$, then $f(g) = f(g')$. But then $g'g^{-1} \in \ker f$, so $g\ker f = g'\ker f$.

■

Example

Consider the homomorphism

$$\begin{aligned}\exp : (\mathbb{C}, +) &\to (\mathbb{C}^*, \times) \\ z &\mapsto e^z\end{aligned}$$

Then $\ker\exp = 2\pi i\mathbb{Z}$, so $\mathbb{C}^* \cong \mathbb{C}/2\pi i\mathbb{Z}$.

Theorem 2.9 (Second isomorphism theorem)

Let $H \leq K$ and $K \lhd G$. Then

$$HK = \{hk : h \in H, k \in K\}$$

is a subgroup of G. Furthermore, $H \cap K$ is normal in H, and there is an isomorphism $HK/K \cong H/H \cap K$.

Proof. We will show the results in order.

- *HK is a subgroup.* It contains e_G, so is non-empty. Now consider $x = hk$ and $y = h'k'$. We'll show $yx^{-1} \in HK$. Observe

$$\begin{aligned}yx^{-1} &= h'k'(hk)^{-1} \\ &= h'k'k^{-1}h^{-1} \\ &= (h'h^{-1})h(k'k^{-1})h^{-1} \\ &= h'h^{-1}k''\end{aligned}$$

  by normality of K.

- *$H \cap K$ is a kernel.* Consider

$$\begin{aligned}\phi : H &\to G/K \\ h &\mapsto hK\end{aligned}$$

  by restricitng the domain from $G \to G/K$ – the kernel is $H \cap K$.

- *Final statement*. By the first isomorphism theorem, we have

$$\operatorname{Im}\phi \cong H/(H \cap K)$$

so we must show $\operatorname{Im}\phi \cong HK/K$. But $\operatorname{Im}\phi$ is the set of K-cosets in G that are representable by elements of H, i.e. $\{hK : h \in H\} \subseteq G/K = HK/K$.

■

§2.1 Correspondence between subgroups of G and G/K

Let $p : G \to G/K$ be the quotient homomorphism. We have a bijection

$$\begin{aligned}\{\text{subgroups of } G \text{ containing } K\} &\leftrightarrow \{\text{subgroups of } G/K\}\\ K \lhd L \leq G &\mapsto L/K \leq G/K\\ \{g \in G : gK \in A\} &\leftarrow\!\shortmid A \leq G/K\end{aligned}$$

The forward bijection is just applying the projection map on a subset, and, since the quotient map is a homomorphism, its image is a subgroup; and the backward bijection is actually just applying the preimage of p. It is an exercise to check that the forward and backward directions are inverses.

> **Remark.** In fact the correspondence matches normal subgroups on the two sides.

Theorem 2.10 (Third isomorphism theorem)
Let $K \leq L \leq G$ be normal subgroups of G. Then there exists an isomorphism $(G/K)/(L/K) \cong G/L$.

Proof. Define a map

$$\begin{aligned}\phi : G/K &\to G/L\\ gK &\mapsto gL\end{aligned}$$

- ϕ *is well-defined*. If $gK = g'K$, then $g'g^{-1} \in K \subseteq L \implies gL = g'L$.
- ϕ *is a homomorphism*. Clear.

We have $\ker\phi = \{gK : gL = L\} = \{gK : g \in L\} = L/K$. Hence we are done by the first isomorphism theorem. ■

Definition 2.11 (Simplicity)**.** A group G is called *simple* if its only normal subgroups are G and $\{e\}$.

Proposition 2.12
Let G be an abelian group. Then G is simple if and only if $G \cong C_p$ for some prime p.

Proof. If $G \cong C_p$, then any non-identity element generates G by, for example, Lagrange. So G is simple.

If G is simple and abelian, then take any $g \neq e \in G$ and consider $\langle g \rangle$. This is a subgroup, and since G is abelian, it is automatically normal. Hence, by simplicity, $\langle g \rangle = G$, and so G is cyclic.

If G is infinite and cyclic, then $G \cong \mathbb{Z}$, but $\mathbb{Z}$ is not simple, so this is a contradiction. Hence G is finite and $G \cong C_m$. If q divides m, take the subgroup generated by $g^{m/q}$, where g generates G. Normality is automatic, so $q = 1$ or $q = m$ by simplicity – in other words, m is prime. ■

Theorem 2.13

Let G be a finite group. Then, there exist subgroups

$$G = H_1 \triangleright H_2 \triangleright H_3 \triangleright \cdots \triangleright H_n = \{e\}$$

such that H_i/H_{i+1} is simple for all i.

Proof. If G is simple, $H_2 = \{e\}$, and we're done. Otherwise, let H_2 be a proper normal subgroup of maximal order in G. We claim G/H_2 is simple. If not, consider the quotient map $\pi : G \to G/H_2$.

If $N \triangleleft G/H_2$ is a proper normal subgroup, by the correspondence theorem, we can find a proper normal subgroup in G containing H_2. But this contradicts the maximality of H_2.

Now we can just iterate – note that $|H_{i+1}| < |H_i|$, so we will eventually terminate. ■

> **Remark.** This sequence is not necessarily unique.

§3 Group actions and permutations

Definition 3.1. Let X be a set. Let $\mathrm{Sym}(X)$ be the set of permutations on X. The identity is the identity function, and the group law is given by composition. If $X = \{1, \ldots, n\}$, then we write $\mathrm{Sym}(X) = S_n$.

- It's convenient to write $\sigma \in S_n$ as a product of disjoint cycles.
- Every $\sigma \in S_n$ is a product of transpositions, and the number of such transpositions gives rise to the *sign* of σ.

  $$\begin{aligned} \mathrm{sign} : S_n &\to \{\pm 1\} \\ \sigma &\mapsto \begin{cases} +1 & \sigma \text{ is a product of an even number of transpositions} \\ -1 & \sigma \text{ ... odd ...} \end{cases} \end{aligned}$$

  sign is a homomorphism, and is surjective when $n \geq 3$.